Can a Buzz Answer That? How Much of What Blind People Ask a Visual Assistant Fits a Vibration Vocabulary

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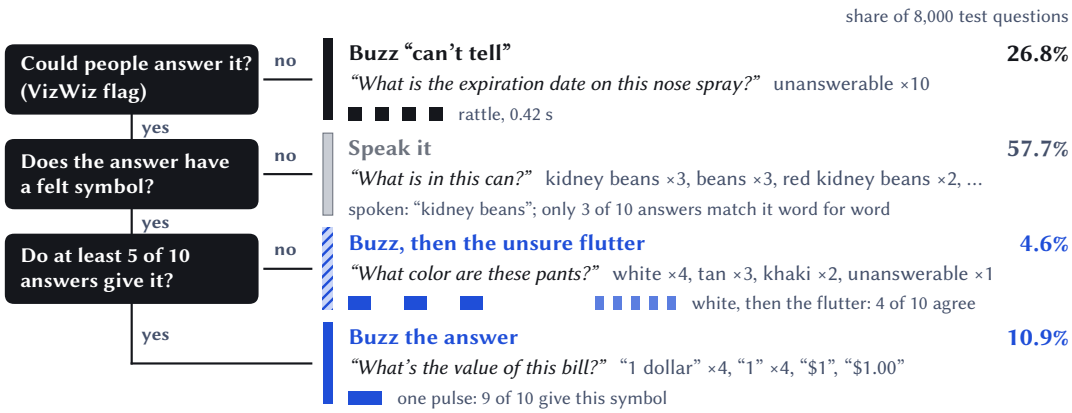


Fig. 1. How each reply is routed. Percentages are shares of the 8,000 held-out test questions. Each route shows a real test question, its crowd answers and the rhythm a phone would play (bars show vibration on, all drawn to one time scale).

Visual assistants answer blind people by speech, occupying the hearing they rely on. Could a vibration rhythm answer instead? For 32,842 VizWiz questions, we routed the crowd's answer to a felt symbol (can't tell, yes, no, number, colour) or speech, fixing all rules before opening 8,000 held-out questions. Rhythms could encode an informative answer for only 15.5% of these (10.9% with at least half agreeing, 4.6% marked unsure) and "can't tell" for 26.8% more. Rhythms were not reliably shorter than estimated speech. An off-the-shelf visual question answering model not trained to abstain confidently buzzed an answer for 20% of photos the crowd marked unanswerable. A felt channel should carry failure and uncertainty first, behind an assistant that can say it cannot tell. Recognition is untested; an open Android prototype is built to play rhythms and run identification tests.

CCS Concepts: • **Human-centered computing** → **Accessibility technologies**; *Haptic devices.*

Additional Key Words and Phrases: blind and low vision, visual question answering, vibrotactile feedback, tactons, uncertainty, VizWiz

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1 Introduction

When blind and low-vision (BLV) people ask what is in a photo they took, a visual assistant answers by voice, whether it is a crowd worker [8, 9] or an AI model [17]. Speech competes with the ambient sounds BLV people use to travel safely [16], and people nearby can hear it [30]. Yet many answers are very short: a colour, a yes, the value of a banknote.

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Such short messages suit vibration patterns, or tactons [11, 12, 29]. Tactile aids for BLV people mostly guide movement or render braille [28, 31, 32], and we found no study of how many real answers a small vibration vocabulary could cover.

Answers are also often uncertain. Crowd workers disagree [7, 19], many photos cannot be answered at all [15, 20], and blind users miss many AI errors and have to find their own ways to check answers [1, 21, 27]. So a vibration channel would need to say "I can't tell" and "I'm not sure" too.

We test how far such a channel could go with 32,842 questions that blind people asked about their own photos, each with 10 crowd answers [20]. We contribute (1) an estimate of which replies fit a rhythm vocabulary of can't tell, yes, no, numbers and colours (Fig. 1), and of how consistently people give them; (2) a check with a real visual question answering (VQA) model, showing why the channel needs a failure signal the model can produce; and (3) an open prototype, Felt Answers, for testing the rhythms with people.

2 Related Work

What blind people ask. Most VizWiz questions ask to identify an object, describe it (often its colour) or read text [8, 9]. Yes/no and number answers are rare, and a quarter to a third of the questions cannot be answered from the photo [20]. Later work labelled why answers differ [7], what is wrong with the photos [15] and which vision skills a question needs [35]. BLV people now use multimodal AI assistants every day and run into made-up answers [17, 22, 34].

Touch. Tactons encode messages in rhythm, the parameter people identify most reliably, and in roughness [11, 12]. Counted pulses can convey digits on the wrist [29], and smartwatches already tap out the time [4]. People identify a handful of vibrotactile cues without training, and more with practice [5, 14, 26]. Vibration has rendered braille [23, 28] and colour [13], while camera aids guide aiming mostly with speech or tones [24, 32].

3 Method

Data. We used all VizWiz-VQA questions: training (20,523), validation (4,319) and test (8,000), whose 328,420 crowd answers are now public. For training and validation we also used labels for why answers differ [7] and for image-quality flaws [15], counting a label if at least 2 of 5 annotators chose it. Answers were normalised with the official VQA rules [3].

The assistant's answer. We wanted to study how an answer is delivered, not whether it is right, so the assistant says what the crowd says: "can't tell" if VizWiz marks the question unanswerable, and otherwise the most frequent answer. This bounds how correct a vibration reply could be, but not how many questions could get one.

Vocabulary. All rules were set on the training and validation splits before the test split was opened (Table 1). Tier 0 is meant to need little learning: can't tell, yes, no, and numbers up to 999, played digit by digit as counted pulses. Tier 1 adds the 11 basic colour terms [6], one or two per answer and without modifiers ("light blue" is spoken), as rhythms of 3 or 4 elements, shorter for frequent colours. Tier 2 adds K learned symbols for other frequent answers. The categories set coverage; the rhythms set only duration and which codes may be confused. Pulses of 100–600 ms with gaps of at least 150 ms suit phone actuators [2] and counting [29]. Two signals break this rule on purpose: the can't-tell rattle and the unsure flutter are fast pulse trains meant to feel rough, not countable [12].

Routing. An answer without a symbol is spoken. An answer with one (a felt answer) is played alone if at least half of the 10 crowd answers map to that symbol (agreement $k \geq 5$), and is otherwise followed by the flutter. Can't tell is never marked unsure. We estimated speech time from word counts at 180 words per minute, the top of typical speaking rates, and at 334, the mean maximum intelligible rate for visually impaired listeners [10]. Both rates favour speech. Neither estimate includes latency: decoding and motor time for rhythms, synthesis for speech. In speech, unsure answers start with "I'm not sure, but", and can't tell is "I can't tell, please take another photo" or just "Can't tell".

Table 1. The vibration vocabulary. s: 100 ms pulse; l: 300 ms pulse; elements 150 ms apart; symbols 500 ms apart. Examples are drawn to one time scale (bars show vibration on): the number is 3, the colour is green, and unsure shows the flutter alone.

Meaning	Rhythm	Example	Duration
Can't tell	rattle: four 60 ms pulses, 60 ms apart		0.42 s
Yes; no	l; s s		0.30; 0.35 s
Number	digits as counted 150 ms pulses (200 ms gaps, 450 after five), 600 ms between digits; 0: one 600 ms pulse		0.15–10.8 s
Colour	white sss, grey ssl, black sls, blue lss, red sll, brown lsl, pink lls, green lll, purple ssss, yellow sssl, orange ssls		0.60–2.75 s
Unsure	flutter after the answer: five 40 ms pulses, 40 ms apart		+0.86 s

A real model. Real assistants can be wrong, so we ran ViLT-B/32 fine-tuned on VQAv2 [18, 25] zero-shot on all 8,000 test questions. ViLT picks one of 3,129 answers, none of which is "unanswerable". Its confidence p is the sigmoid of the top logit, as it was trained on soft VQA scores; with the same symbol rules, a confident buzz ($p \geq 0.5$) is played alone and others with the flutter. A buzz on an answerable photo counts as right if at least 3 of the 10 crowd answers map to its symbol. The analysis plan was fixed before inference, with one later addendum. ViLT is older and weaker than current assistants and trained neither on VizWiz nor to abstain (VQA accuracy 17% here), but it is open and runs offline.

Prototype. Felt Answers is an offline web page for Chrome on Android with four modes. *Ask* plays 228 real test questions as rhythm or speech, *Learn* lists every code, *Test* runs identification trials in shuffled blocks with a confusion matrix, and *Check* lets a tester rate each pattern on a table and in the hand and logs vibration timing. Results export as CSV or JSON, and nothing is sent online. Its JavaScript routing matched the Python analysis on all 32,842 questions.

4 Results

A buzz could cover 42.3% of replies, but only 15.5% would be informative answers. Of the 8,000 test questions, 26.8% were marked unanswerable and would get can't tell, and 15.5% got a felt answer (colour 8.6%, no 2.6%, yes 2.4%, number 1.9%), 10.9% without the flutter and 4.6% with it (Fig. 1). Without colours, felt answers fell to 6.8%. Learned symbols helped slowly against a long tail of 6,397 distinct training answers: even 128 symbols, far more than people identify without training [5, 14], raised felt replies only to 55.5%, although felt answers nearly doubled, to 28.6% (Fig. 2a).

Closed answers agreed more, partly by construction. Among 17,925 answerable training and validation questions, at least half of the answerers gave the same symbol for 79% of yes, 75% of no, 69% of number and 66% of colour answers. For spoken answers, at least half gave the same string only 42% of the time (Fig. 2b). The comparison favours symbols, since open answers can differ in wording without differing in meaning [7]. On the test set, 29% of felt answers fell below $k = 5$ and carried the flutter. Even can't tell was contested: for 12% of can't-tell questions (261 of 2,147), only 4 or 5 of the 10 answerers said the question was unanswerable.

A model not trained to abstain buzzed on many unanswerable photos. ViLT buzzed on 35.3% of test questions, 23.3% confidently. On the 2,147 photos the crowd marked unanswerable it buzzed 38% of the time, 20% confidently, and gave an abstention-like answer such as "unknown" for only 5% of them (unplanned; such answers are spoken). Its confidence barely separated unanswerable from answerable photos (AUROC 0.57, also unplanned), but it did carry some signal: on answerable photos, confident buzzes were right 41% of the time and flutter-marked ones 6%. Of the 1,867 confident buzzes, 68% were wrong: fewer than 3 of 10 answerers shared the symbol, or the photo was unanswerable (Fig. 2c). Thresholds of 0.3 or 0.7, softmax confidence or a looser definition of right gave 64–72%.

Rhythms were not reliably shorter than speech. At 180 words per minute the rhythm was shorter than speech for 75% of felt replies, mostly because of can't tell: the rattle lasts 0.42 s, against 0.67 s for just the words "can't tell". Yes was only slightly shorter (0.30 s against 0.33 s), and no slightly longer (0.35 s). Against "can't tell", the share fell to 11% if

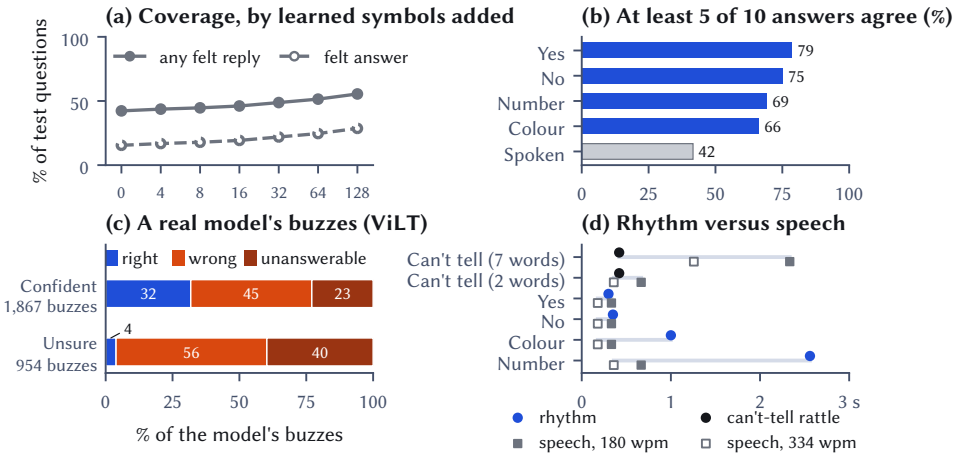


Fig. 2. (a) Test questions with any felt reply (including can't tell) or a felt answer, as learned symbols are added. (b) Answerable training and validation questions where at least 5 of 10 answerers agree (on the symbol, or for spoken answers the exact string). (c) ViLT's test buzzes: right, wrong, or on an unanswerable photo. (d) Median rhythm duration against estimated speech time.

the rattle kept our 100 ms pulse rule (0.85 s), and to 0.4% at 334 words per minute (0.36 s). Colour (1.00 s) and number (2.56 s) rhythms took longer to play than to say (Fig. 2d). The stronger case for touch is the free ear: counting only can't tell and confident felt answers, touch would have kept hearing free for 37.7% of questions.

5 Design Implications

Use touch for failure, doubt and short closed answers. The most frequent felt message is a failure, and without the flutter, touch would overstate confidence, an error blind users find hard to catch [21]. Yes/no answers, numbers and colours are candidates for touch, since answerers agree on them more often and blind users prefer short answers to questions about one attribute [22]. **Keep reading, naming and description in speech.** Most replies (57.7%) had no symbol. **The assistant must be able to abstain.** ViLT rarely did, so trained abstention or photo-quality detection has to sit behind a felt "can't tell" [15, 33]. A retake rhythm could not reliably say why: blur, framing or darkness was flagged for 87% of unanswerable and 67% of answerable labelled photos. **Some codes differ only in timing:** no and two by 50 ms in pulse and gap, yes and one in pulse length. Keyword rules on the question's wording predicted the category of 83% of felt answers, so context may tell them apart; if not, each category could open with its own element.

Limitations. Our model is a weak zero-shot probe. No one has felt these rhythms in a study yet: rendering on phones, identification, learning and preference over speech are untested. VizWiz questions are over a decade old and from one community, speech times are estimates, and Web Vibration works only on Android.

Conclusion. Over a quarter of VizWiz questions could get a felt "can't tell" but few an informative answer, so a vibration channel should start with failure, behind a model that can abstain.

Ethics and use of AI. We used public, de-identified data (CC BY 4.0); there were no participants. An AI assistant (Claude, Anthropic) helped search the literature, wrote and ran the analysis, figure and prototype code under the author's direction, and drafted the text; the author is responsible for the content.

References

[1] Rahaf Alharbi, Pa Lor, Jaylin Herskovitz, Sarita Schoenebeck, and Robin N. Brewer. 2024. Misfitting With AI: How Blind People Verify and Contest AI Errors. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '24)*. ACM, 1–17. doi:10.1145/3663548.3675659

[2] Android Developers. 2026. Haptics Design Principles. https://developer.android.com/develop/ui/views/haptics/haptics-principles Accessed 17 October 2026.

[3] Stanislaw Antol, Aishwarya Agrawal, Jiasen Lu, Margaret Mitchell, Dhruv Batra, C. Lawrence Zitnick, and Devi Parikh. 2015. VQA: Visual Question Answering. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. IEEE, 2425–2433. doi:10.1109/ICCV.2015.279

[4] Apple Inc. 2026. Tell Time with Haptic Feedback on Apple Watch. Apple Watch User Guide. https://support.apple.com/guide/watch/apd2c755c294/watchos Accessed 17 October 2026.

[5] Mojtaba Azadi and Lynette A. Jones. 2014. Evaluating Vibrotactile Dimensions for the Design of Tactons. *IEEE Transactions on Haptics* 7, 1 (2014), 14–23. doi:10.1109/TOH.2013.2296051

[6] Brent Berlin and Paul Kay. 1969. *Basic Color Terms: Their Universality and Evolution*. University of California Press, Berkeley, CA.

[7] Nilavra Bhattacharya, Qing Li, and Danna Gurari. 2019. Why Does a Visual Question Have Different Answers?. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 4270–4279. doi:10.1109/ICCV.2019.00437

[8] Jeffrey P. Bigham, Chandrika Jayant, Hanjie Ji, Greg Little, Andrew Miller, Robert C. Miller, Robin Miller, Aubrey Tatarowicz, Brandyn White, Samual White, and Tom Yeh. 2010. VizWiz: Nearly Real-Time Answers to Visual Questions. In *Proceedings of the 23rd Annual ACM Symposium on User Interface Software and Technology (UIST '10)*. ACM, 333–342. doi:10.1145/1866029.1866080

[9] Erin Brady, Meredith Ringel Morris, Yu Zhong, Samuel White, and Jeffrey P. Bigham. 2013. Visual Challenges in the Everyday Lives of Blind People. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '13)*. ACM, 2117–2126. doi:10.1145/2470654.2481291

[10] Danielle Bragg, Cynthia Bennett, Katharina Reinecke, and Richard Ladner. 2018. A Large Inclusive Study of Human Listening Rates. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. ACM, 1–12. doi:10.1145/3173574.3174018

[11] Stephen Brewster and Lorna M. Brown. 2004. Tactons: Structured Tactile Messages for Non-Visual Information Display. In *Proceedings of the Fifth Australasian User Interface Conference (AUIC '04, CRPIT 28)*. Australian Computer Society, 15–23.

[12] Lorna M. Brown, Stephen A. Brewster, and Helen C. Purchase. 2005. A First Investigation into the Effectiveness of Tactons. In *First Joint Eurohaptics Conference and Symposium on Haptic Interfaces for Virtual Environment and Teleoperator Systems (WHC '05)*. IEEE, 167–176. doi:10.1109/WHC.2005.6

[13] Marta G. Carcedo, Soon Hau Chua, Simon Perrault, Paweł Woźniak, Raj Joshi, Mohammad Obaid, Morten Fjeld, and Shengdong Zhao. 2016. HaptiColor: Interpolating Color Information as Haptic Feedback to Assist the Colorblind. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)*. ACM, 3572–3583. doi:10.1145/2858036.2858220

[14] Elyse D. Z. Chase, Ali Israr, Pornthep Preechayasomboon, Sarah Sykes, Aakar Gupta, and Jessica Hartcher-O'Brien. 2021. Learning Vibes: Communication Bandwidth of a Single Wrist-Worn Vibrotactile Actuator. In *2021 IEEE World Haptics Conference (WHC)*. IEEE, 421–426. doi:10.1109/WHC49131.2021.9517208

[15] Tai-Yin Chiu, Yinan Zhao, and Danna Gurari. 2020. Assessing Image Quality Issues for Real-World Problems. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 3643–3653. doi:10.1109/CVPR42600.2020.00370

[16] Nicholas A. Giudice and Gordon E. Legge. 2008. Blind Navigation and the Role of Technology. In *The Engineering Handbook of Smart Technology for Aging, Disability, and Independence*, Abdelsalam Helal, Mounir Mokhtari, and Bessam Abdulrazak (Eds.). Wiley, 479–500. doi:10.1002/9780470379424.ch25

[17] Ricardo E. Gonzalez Penuela, Jazmin Collins, Cynthia L. Bennett, and Shiri Azenkot. 2024. Investigating Use Cases of AI-Powered Scene Description Applications for Blind and Low Vision People. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, 1–21. doi:10.1145/3613904.3642211

[18] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. 2017. Making the V in VQA Matter: Elevating the Role of Image Understanding in Visual Question Answering. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 6325–6334. doi:10.1109/CVPR.2017.670

[19] Danna Gurari and Kristen Grauman. 2017. CrowdVerge: Predicting If People Will Agree on the Answer to a Visual Question. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17)*. ACM, 3511–3522. doi:10.1145/3025453.3025781

[20] Danna Gurari, Qing Li, Abigale J. Stangl, Anhong Guo, Chi Lin, Kristen Grauman, Jiebo Luo, and Jeffrey P. Bigham. 2018. VizWiz Grand Challenge: Answering Visual Questions from Blind People. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 3608–3617. doi:10.1109/CVPR.2018.00380

[21] Jonggi Hong and Hernisa Kacorri. 2024. Understanding How Blind Users Handle Object Recognition Errors: Strategies and Challenges. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '24)*. ACM, 1–15. doi:10.1145/3663548.3675635

[22] Mina Huh, Fangyuan Xu, Yi-Hao Peng, Chongyan Chen, Hansika Murugu, Danna Gurari, Eunsol Choi, and Amy Pavel. 2024. Long-Form Answers to Visual Questions from Blind and Low Vision People. In *Proceedings of the Conference on Language Modeling (COLM)*. arXiv:2408.06303

[23] Chandrika Jayant, Christine Acuario, William Johnson, Janet Hollier, and Richard Ladner. 2010. V-Braille: Haptic Braille Perception Using a Touch-Screen and Vibration on Mobile Phones. In *Proceedings of the 12th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '10)*. ACM, 295–296. doi:10.1145/1878803.1878878

[24] Chandrika Jayant, Hanjie Ji, Samuel White, and Jeffrey P. Bigham. 2011. Supporting Blind Photography. In *Proceedings of the 13th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '11)*. ACM, 203–210. doi:10.1145/2049536.2049573

[25] Wonjae Kim, Bokyung Son, and Ildoo Kim. 2021. ViLT: Vision-and-Language Transformer Without Convolution or Region Supervision. In *Proceedings of the 38th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 139)*. PMLR, 5583–5594.

[26] Seungyon Claire Lee and Thad Starner. 2010. BuzzWear: Alert Perception in Wearable Tactile Displays on the Wrist. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '10)*. ACM, 433–442. doi:10.1145/1753326.1753392

[27] Haley MacLeod, Cynthia L. Bennett, Meredith Ringel Morris, and Edward Cutrell. 2017. Understanding Blind People's Experiences with Computer-Generated Captions of Social Media Images. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17)*. ACM, 5988–5999. doi:10.1145/3025453.3025814

[28] Hugo Nicolau, João Guerreiro, Tiago Guerreiro, and Luís Carriço. 2013. UbiBraille: Designing and Evaluating a Vibrotactile Braille-Reading Device. In *Proceedings of the 15th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '13)*. ACM, 1–8. doi:10.1145/2513383.2513437

[29] Jerome Pasquero, Scott J. Stobbe, and Noel Stonehouse. 2011. A Haptic Wristwatch for Eyes-Free Interactions. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11)*. ACM, 3257–3266. doi:10.1145/1978942.1979425

[30] Kristen Shinohara and Jacob O. Wobbrock. 2011. In the Shadow of Misperception: Assistive Technology Use and Social Interactions. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11)*. ACM, 705–714. doi:10.1145/1978942.1979044

[31] Lee Stearns, Ruofei Du, Uran Oh, Catherine Jou, Leah Findlater, David A. Ross, and Jon E. Froehlich. 2016. Evaluating Haptic and Auditory Directional Guidance to Assist Blind People in Reading Printed Text Using Finger-Mounted Cameras. *ACM Transactions on Accessible Computing* 9, 1 (2016), 1–38. doi:10.1145/2914793

[32] Marynel Vázquez and Aaron Steinfeld. 2014. An Assisted Photography Framework to Help Visually Impaired Users Properly Aim a Camera. *ACM Transactions on Computer-Human Interaction* 21, 5, Article 25 (2014). doi:10.1145/2651380

[33] Spencer Whitehead, Suzanne Petryk, Vedaad Shakib, Joseph Gonzalez, Trevor Darrell, Anna Rohrbach, and Marcus Rohrbach. 2022. Reliable Visual Question Answering: Abstain Rather Than Answer Incorrectly. In *Computer Vision – ECCV 2022 (Lecture Notes in Computer Science, Vol. 13696)*. Springer, 148–166. doi:10.1007/978-3-031-20059-5_9

[34] Jingyi Xie, Rui Yu, He Zhang, Syed Masum Billah, Sooyeon Lee, and John M. Carroll. 2025. Beyond Visual Perception: Insights from Smartphone Interaction of Visually Impaired Users with Large Multimodal Models. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, 1–17. doi:10.1145/3706598.3714210

[35] Xiaoyu Zeng, Yanan Wang, Tai-Yin Chiu, Nilavra Bhattacharya, and Danna Gurari. 2020. Vision Skills Needed to Answer Visual Questions. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW2 (2020), 1–31. doi:10.1145/3415220